Hospital Homelab Project - Gantt Chart Timeline

Spring 2026 Semester (January 28 - April 7, 2026)

Timeline Overview (10 Weeks)

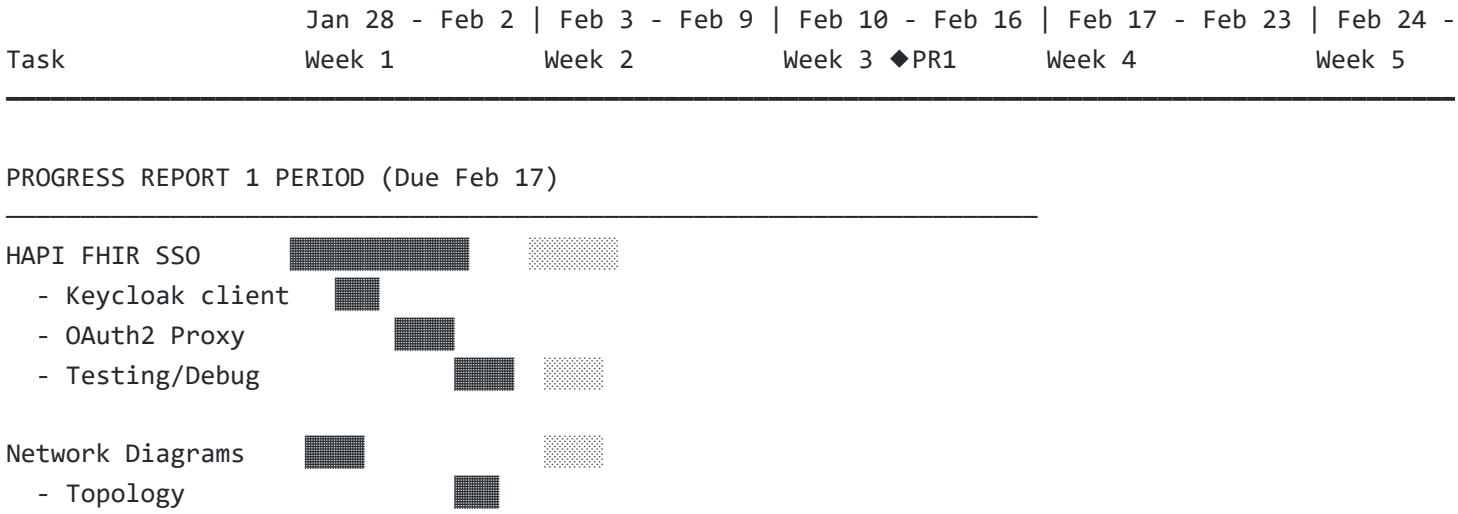
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Week 1-3: Progress Report 1 Period (Jan 28 - Feb 16)
Week 4-6: Progress Report 2 Period (Feb 17 - Mar 9)
Week 7-10: Final Expo Preparation (Mar 10 - Apr 7)
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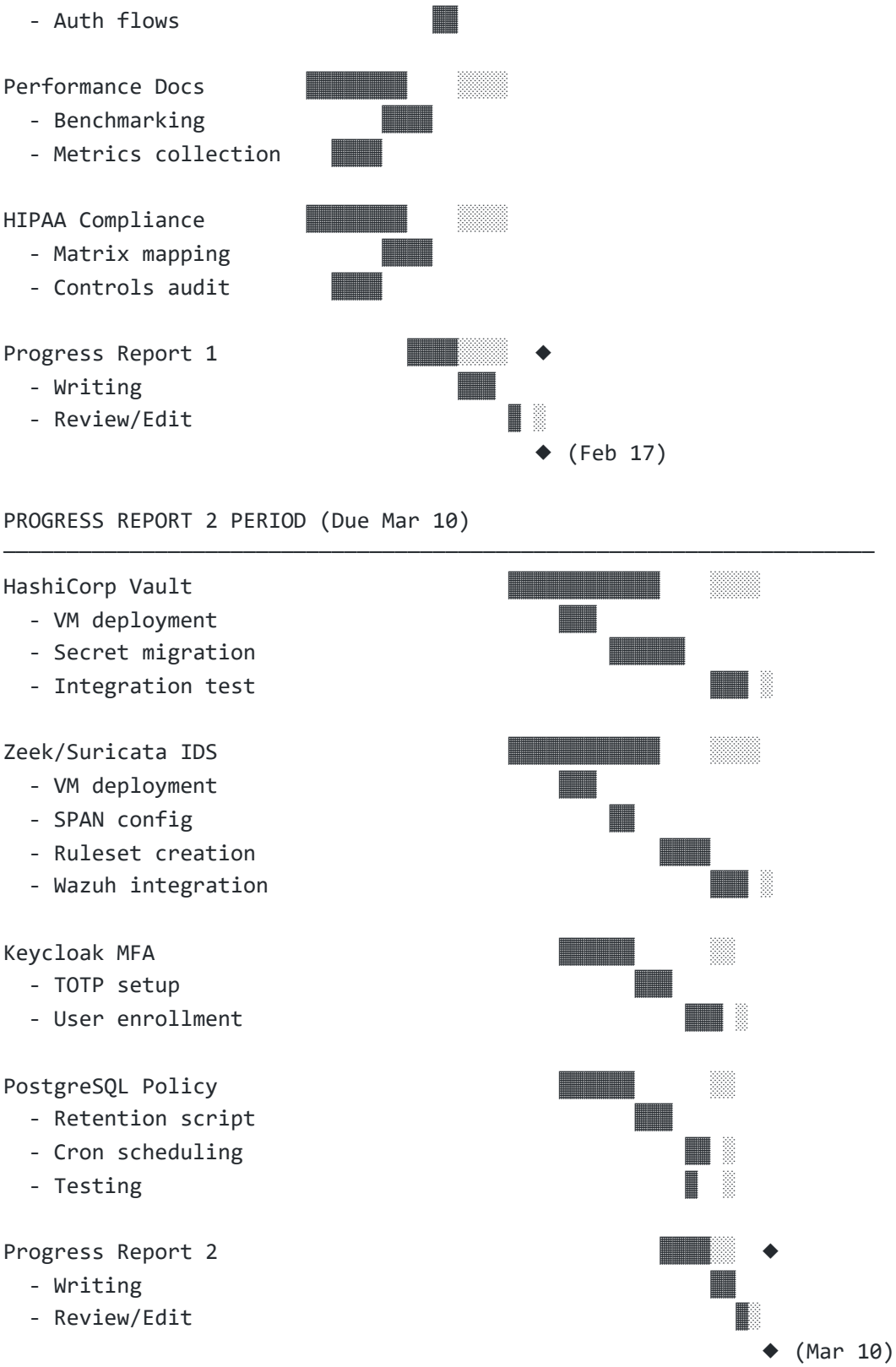
Gantt Chart - Task Timeline

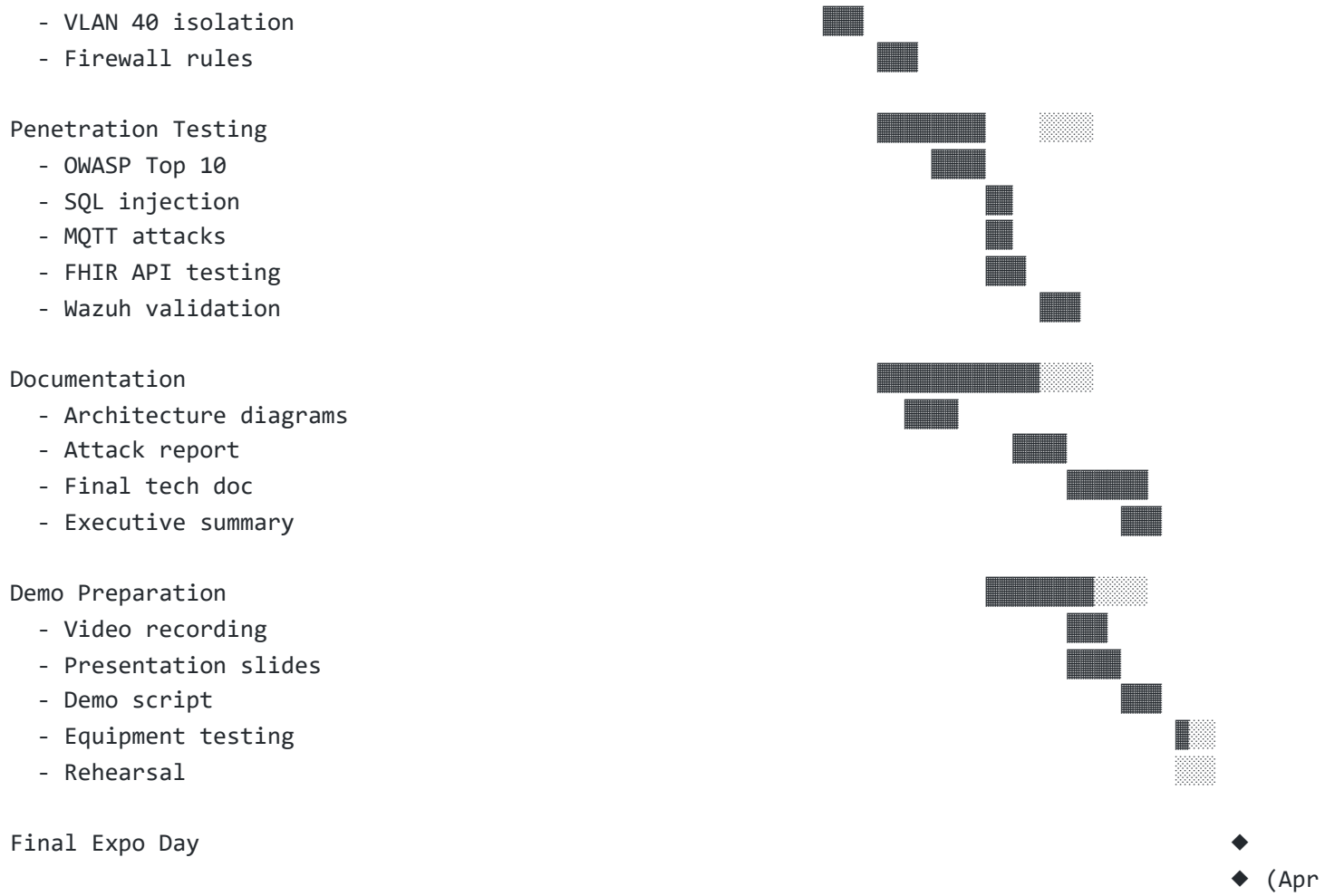
Legend

- ████ Completed
- ▓▓▓▓ In Progress / Planned
- ░░░░ Buffer / Testing Time
- ◆ Milestone/Deadline

Week-by-Week Breakdown




- Auth flows
Performance Docs
- Benchmarking
- Metrics collection
HIPAA Compliance
- Matrix mapping
- Controls audit
Progress Report 1
- Writing
- Review/Edit
(Feb 17)
PROGRESS REPORT 2 PERIOD (Due Mar 10)
HashiCorp Vault
- VM deployment
- Secret migration
- Integration test
Zeek/Suricata IDS
- VM deployment
- SPAN config
- Ruleset creation
- Wazuh integration
Keycloak MFA
- TOTP setup
- User enrollment
PostgreSQL Policy
- Retention script
- Cron scheduling
- Testing
Progress Report 2
- Writing
- Review/Edit
(Mar 10)
FINAL EXPO PREPARATION (Due Apr 7)
Attack Lab Setup
- Kali Linux #1
- Kali Linux #2
- Metasploitable
- T-Pot Honeypot



Detailed Task Breakdown by Milestone

◆ Progress Report 1 (February 17, 2026)

Week 1 (Jan 28 - Feb 2): HAPI FHIR SSO & Documentation Start

- Days 1-2: Create Keycloak client for HAPI FHIR (hapi-fhir-oauth2)
- Days 3-5: Install and configure OAuth2 Proxy on HAPI FHIR VM
- Days 5-7: Begin network topology diagrams and performance benchmarking

Week 2 (Feb 3 - Feb 9): Integration & Metrics

- Days 1-3: Complete OAuth2 Proxy configuration and testing
- Days 4-5: Finalize network diagrams (topology, authentication flows)
- Days 6-7: Collect performance metrics (query times, throughput, latency)

Week 3 (Feb 10 - Feb 16): HIPAA Compliance & Report Writing

- Days 1-3: Create HIPAA compliance matrix (map 164.312 controls to implementations)
- Days 4-5: Write Progress Report 1 narrative
- Days 6-7: Review, edit, and submit Progress Report 1

Deliverables:

- ✓ HAPI FHIR OAuth2 Proxy operational
- ✓ Network topology diagrams (Visio/draw.io)
- ✓ Performance benchmark report
- ✓ HIPAA compliance matrix
- ✓ Progress Report 1 document (PDF)

◆ Progress Report 2 (March 10, 2026)

Week 4 (Feb 17 - Feb 23): HashiCorp Vault & Zeek/Suricata Deployment

- Days 1-2: Deploy HashiCorp Vault VM (10.30.0.58) on TrueNAS
- Days 3-5: Deploy Zeek/Suricata IDS/IPS VM (10.30.0.59)
- Days 6-7: Configure SPAN port on UniFi switch for traffic mirroring

Week 5 (Feb 24 - Mar 2): Secrets Migration & IDS Rulesets

- Days 1-3: Migrate credentials to Vault (PostgreSQL, Keycloak, MQTT broker)
- Days 4-5: Create Zeek/Suricata rulesets for HL7/FHIR protocol anomalies
- Days 6-7: Configure Keycloak MFA (TOTP authenticator)

Week 6 (Mar 3 - Mar 9): Integration Testing & Report Writing

- Days 1-2: Integrate Zeek/Suricata alerts with Wazuh SIEM
- Days 3-4: Implement PostgreSQL data retention policy (90-day rolling window)
- Days 5-7: Write, review, and submit Progress Report 2

Deliverables:

- ✓ HashiCorp Vault operational with secret rotation
- ✓ Zeek/Suricata IDS/IPS with custom rulesets
- ✓ Keycloak MFA enabled for all users
- ✓ PostgreSQL retention policy automated
- ✓ Progress Report 2 document (PDF)

◆ Final Expo (April 7, 2026)

Week 7 (Mar 10 - Mar 16): Attack Lab Infrastructure

- Days 1-2: Deploy Kali Linux VM #1 (external attacker perspective)
- Days 3-4: Deploy Kali Linux VM #2 (internal threat simulation)
- Days 5-6: Deploy Metasploitable 3 and T-Pot Honeypot VMs
- Day 7: Configure VLAN 40 isolation and firewall rules

Week 8 (Mar 17 - Mar 23): Attack Lab Configuration & Initial Testing

- Days 1-2: Configure attack tools on Kali Linux instances
- Days 3-5: Metasploitable configuration and vulnerability verification
- Days 6-7: T-Pot honeypot configuration and alert testing

Week 9 (Mar 24 - Mar 30): Penetration Testing Execution

- Days 1-2: OWASP Top 10 attacks against FHIR API (SQL injection, XSS, auth bypass)
- Days 3-4: MQTT broker authentication bypass attempts
- Days 5-6: Network-level attacks (port scanning, man-in-the-middle)
- Day 7: Wazuh SIEM detection validation and attack documentation

Week 10 (Mar 31 - Apr 6): Final Documentation & Demo Preparation

- Days 1-2: Create system architecture diagrams and attack flow diagrams
- Days 3-4: Write penetration testing report and final technical documentation
- Days 5-6: Record backup demo videos and create presentation slides
- Day 7: Equipment testing, demo rehearsal, and final preparation

Day of Expo (April 7, 2026):

- Morning: Setup booth, test all equipment
- Afternoon: Live demonstrations and Q&A with faculty/industry partners

Deliverables:

- ✓ Fully operational attack lab (4 VMs on isolated subnet)
- ✓ Penetration testing report with OWASP Top 10 coverage
- ✓ Final technical documentation (30+ page report)
- ✓ System architecture diagrams (network, data flow, authentication)
- ✓ Presentation slides (15-minute talk)
- ✓ Backup demo videos (3-5 minutes each)

- ✓ Live demonstration script with fallback plans

Critical Path Analysis

Must-Complete Tasks (Cannot be delayed):

1. HAPI FHIR OAuth2 Proxy (Week 1-2) → Blocks PR1
2. Progress Report 1 writing (Week 3) → Hard deadline Feb 17
3. Vault/IDS deployment (Week 4-5) → Blocks PR2
4. Progress Report 2 writing (Week 6) → Hard deadline Mar 10
5. Attack lab deployment (Week 7-8) → Blocks penetration testing
6. Final documentation (Week 10) → Hard deadline Apr 7

Flexible Tasks (Can be descoped if time constrained):

- T-Pot Honeypot → Nice-to-have, not critical for demo
- PostgreSQL retention policy → Can document as "future work"
- Kali Linux #2 → One Kali instance sufficient for demo
- MFA configuration → Can demonstrate capability without full rollout

Parallel Execution Opportunities:

- Week 4: Vault deployment + IDS deployment (independent tasks)
- Week 7: Kali #1 + Kali #2 + Metasploitable (can deploy simultaneously)
- Week 10: Documentation + Demo video recording (different work streams)

Resource Allocation & Time Budget

Milestone	Total Hours	Breakdown
Progress Report 1	20 hours	8h implementation, 6h documentation, 6h report writing
Progress Report 2	30 hours	14h implementation, 8h testing, 8h report writing

Milestone	Total Hours	Breakdown
Final Expo Prep	50 hours	16h attack lab, 12h pentesting, 12h documentation, 10h demo prep
Total Semester	**100 hours**	~10 hours/week average

Weekly Time Commitment:

- Weeks 1-3: 6-8 hours/week (PR1 period, lighter workload)
- Weeks 4-6: 10-12 hours/week (PR2 period, VM deployments)
- Weeks 7-10: 12-15 hours/week (Expo prep, heavy documentation)

Contingency Plans

If Behind Schedule:

1. **By Feb 10** (Week 3): Skip OAuth2 Proxy, document HAPI FHIR SSO as "future work" in PR1
2. **By Feb 24** (Week 5): Deploy only Vault OR IDS/IPS (not both) for PR2
3. **By Mar 17** (Week 8): Reduce attack lab to 1 Kali + Metasploitable (skip T-Pot)
4. **By Mar 31** (Week 10): Simplify penetration testing to 3-4 OWASP attacks instead of full Top 10

If Ahead of Schedule:

1. **By Feb 10** (Week 3): Add Orthanc PACS VM for medical imaging (DICOM)
2. **By Feb 24** (Week 5): Implement email/Slack alerting for Grafana/Wazuh
3. **By Mar 17** (Week 8): Add STIG compliance scanning with OpenSCAP
4. **By Mar 31** (Week 10): Create interactive Jupyter notebook for data analysis demos

Risk Mitigation Timeline

Week	Primary Risk	Mitigation Action
Week 1-2	OAuth2 Proxy complexity	Allocate 2 days for troubleshooting, have skip plan ready
Week 4-5	VM resource exhaustion	Monitor TrueNAS CPU/RAM, prepare external hardware if needed

Week	Primary Risk	Mitigation Action
Week 7-8	Attack lab network instability	Deploy VLAN 40 early, test isolation before full deployment
Week 9	Penetration testing fails to trigger alerts	Review Wazuh rules proactively, adjust sensitivity
Week 10	Demo hardware failure	Record backup videos by Apr 4, test equipment Apr 5-6

Milestones & Deadlines Summary

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◆ Feb 17, 2026 - Progress Report 1 Due
  └─ 3 weeks from today (Jan 28)
  └─ Deliverables: HAPI FHIR SSO, diagrams, HIPAA matrix, PR1 document

◆ Mar 10, 2026 - Progress Report 2 Due
  └─ 6 weeks from today (3 weeks after PR1)
  └─ Deliverables: Vault, IDS/IPS, MFA, retention policy, PR2 document

◆ Apr 7, 2026 - Final Expo & Documentation Due
  └─ 10 weeks from today (4 weeks after PR2)
  └─ Deliverables: Attack lab, pentest report, final docs, presentation, live demo

◆ Apr 7, 2026 - Semester Ends
  └─ All work must be completed by this date
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END OF GANTT CHART

Quick Reference: Key Dates

- **Today:** January 28, 2026 (Week 1, Day 1)
- **PR1 Due:** February 17, 2026 (Week 3, Day 7) - 20 days away
- **PR2 Due:** March 10, 2026 (Week 6, Day 7) - 41 days away
- **Final Expo:** April 7, 2026 (Week 10, Day 7) - 69 days away
- **Days Remaining:** 69 days total, 10 weeks, 2.5 months